QUALITY INFORMATION DOCUMENT

For
OMI_CLIMATE_OFC_BALTIC_area_averaged_anomalies

Issue: 2.0

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RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
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1) INTRODUCTION

a) Product Name

OMI_CLIMATE_OFC_BALTIC_area_averaged_anomalies.

b) Indicator Evaluation Method

Ocean freshwater content anomaly (OFC) is defined here, according to Boyer et al. (2007), as the salinity (S) deviation from a reference period (1993-2014) and is closely proportional to the average salinity change from z1 = 0 m to z2 = 300 m depth:

$$OFC = \rho(S_{ref}, T_{ref}, p) / \rho(0, T_{ref}, p) \cdot (S_{ref} - S) / S_{ref}$$

with a spatially varying density ρ, computed based on potential temperature, salinity and pressure using the TEOS10 (IOC et al., 2010).
Integrated time series for ocean freshwater content and their deviation bars are calculated for the Baltic Sea.

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
BALTICSEA_MULTIYEAR_PHY_003_011 (model reanalysis)	QUID: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-BAL-QUID-003-011.pdf PUM: https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-BAL-PUM-003-011_012.pdf

2) SCIENTIFIC VALIDATION

a) Baltic Sea Ocean Freshwater Content

Climate warming has resulted in the intensification of the global hydrological cycle but not necessarily on the regional scale (Pratap and Markonis, 2022). The increase of net precipitation over land and sea areas, decrease of ice cover, and increase of river runoff are the main components of the global hydrological cycle that increase freshwater content in the ocean (Boyer et al., 2007) and decrease ocean salinity.
The method for calculating the ocean freshwater content anomaly is based on the daily mean sea water salinity fields (*S*) derived from the Baltic Sea reanalysis product BALTICSEA_MULTIYEAR_PHY_003_011. The total freshwater content is determined using the following formula:

$$OFC(t) = \iiint v\, OFC(x, y, z, t)\, dx\, dy\, dz$$

The vertical integral is computed using the static cell vertical thicknesses sourced from the reanalysis product BALTICSEA_MULTIYEAR_PHY_003_011 dataset cmems_mod_bal_phy_my_static, spanning from the sea surface to the ocean bottom. Spatial integration is performed over the Baltic Sea spatial domain, defined as the region between 9° - 31° E and 53° - 66° N.

The Baltic Sea is one of the marginal seas where water salinity and OFC are strongly influenced by the water exchange with the North Sea. The Major Baltic Inflows (MBIs) are the most voluminous event-type sources of saline water to the Baltic Sea (Mohrholz, 2018). The frequency and intensity of the MBIs and other large volume inflows have no long-term trends but do have a multidecadal variability of about 30 years (Mohrholz, 2018; Lehmann and Post, 2015; Lehmann et al., 2017; Radtke et al., 2020). Smaller barotropic and baroclinically driven inflows transport saline water into the halocline or below it, depending on the density of the inflow water (Reissmann et al., 2009).
The Baltic Sea's ocean freshwater content is exhibiting a declining trend of -37±8.8 km³/year (see Figure), along with decadal fluctuations as also noted by Lehmann et al. (2022). Elevated freshwater levels were recorded prior to the Major Baltic Inflows of 1993, 2002, and 2013, which subsequently led to a swift decrease in freshwater content. The lowest ocean freshwater content was recorded in 2019. Over the past four years, the freshwater content anomaly has remained comparatively stable.

Linear trend of -38.1±8.7 km³/year has been estimated from the annual OFC values with the uncertainty of 1.96-times standard error. Interannual changes of FWC were explained by large volume saline water inflows, net precipitation and zonal wind stress. We evaluate the uncertainty from the mean standard deviation of monthly mean OFC. The shaded area in the figure corresponds to the annual standard deviation of monthly mean OFC. The uncertainty time series reveals a decreasing trend of −3.3 km³/year ($p<0.05$) in the variability of the Baltic Sea's OFC over time. This reduction in uncertainty from the first to the second half of the time series suggests that the system has become more stable, with less seasonal fluctuation in OFC.

More details and full scientific evaluation can be found in the Copernicus Marine Service Ocean State Report 7 (Raudsepp et al., 2023).

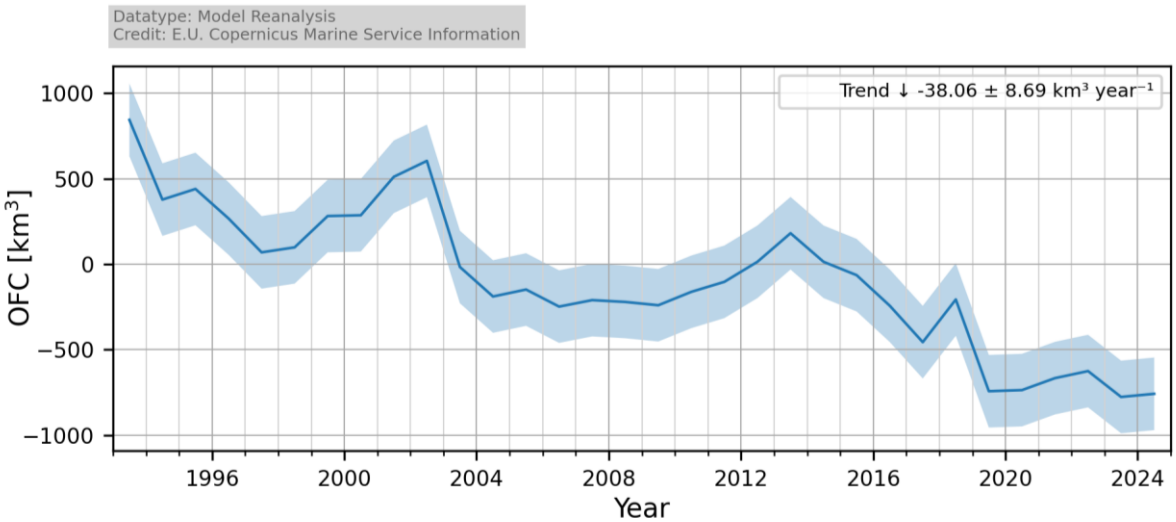

Figure. *The time series of horizontally averaged ocean freshwater content for the period of 1993-2024. The salinity from Copernicus Marine Service regional reanalysis product (*BALTICSEA_MULTIYEAR_PHY_003_011*) have been extracted over the Baltic Sea domain (9 °E - 31 °E; 53 °N - 66 °N).* The shaded area corresponds to the annual standard deviation of monthly mean OFC.

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